Model 2 Writing Brief

Cohort-based dynamic optimisation of investment, water and labour (Task 2)

Source of every number: `Results/v7/model2/` (run of `Code/v7_model2.py`)

2026-11-11

This brief collects what you need to write the Model 2 section: the goal, how the model is built, every formula with its meaning, parameter values, the optimal plan with numbers, the trade-off and sensitivity results, and the decision rules. All money is in CNY at constant prices (M = million). Area in ha, water in m^3, labour in person-days. The plan is a **scenario** for an illustrative 15-year budget, not a surveyed project. Figure files are in `Figures/M2/`.

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1 Goal and what we achieved

Task 2 requirement	How Model 2 meets it	Result
Annual new treated area a_t	Integer number of 50 ha blocks per planting year and irrigation policy	150 ha in year 1, 150 ha in year 2
Annual ecological water w_t	Gross water for every stand still in service, from its lifetime policy, plus PV cleaning; checked yearly and monthly	0.15–1.05 M m^3/yr (Table 6)
Allocation among construction / O&M / revenue facilities (PV)	Three spending streams each year; shares follow from the plan	e.g. year 1: 0.66 / 0.03 / 0.30
Labour input l_t	Person-days = work content of planting, upkeep and PV, capped per year	peak 5,522 (year 2)
Dynamic optimisation built on Task 1	Each planting cohort carries its own Model 1 state (V, M) year by year	3 cohorts tracked 15 years
Explicit ecology–cost trade-off	ε-constraint: minimise net present cost subject to an explicit ecological target	6 points from 0% to 100% of the maximum
Given total budget and horizon	80 M total, 12 M/yr, $T = 15$ years	68.93 M spent
Sensitivity of investment structure (PV, O&M, . . .)	21 one-at-a-time cases at a fixed ecological target	Table 9
Actionable process criterion	PV break-even price rule, water schedule, timing and monitoring rules	Section 7

Table 1: Task 2 requirements and how they are met.

Main results in one paragraph. With an 80 M budget over 15 years, the most ecology the resources allow is 2,097 effective ha·yr (900 ha planted, no PV). Asking for 60% of that (1,258 effective ha·yr), the cheapest plan plants 300 ha in years 1–2 and builds 9.5 MW of PV in years 1–4. It spends 68.93 M (15.75 M planting, 24.68 M upkeep and water, 28.50 M PV) and its net present cost after PV income is 25.35 M; PV saves 2.49 M (9%) compared with the best plan without PV. Each extra effective ha·yr costs about 23 k CNY at low targets and 57 k CNY near the maximum. The tightest resource is the May water cap (95% used), not the annual water total (70%).

2 Model construction

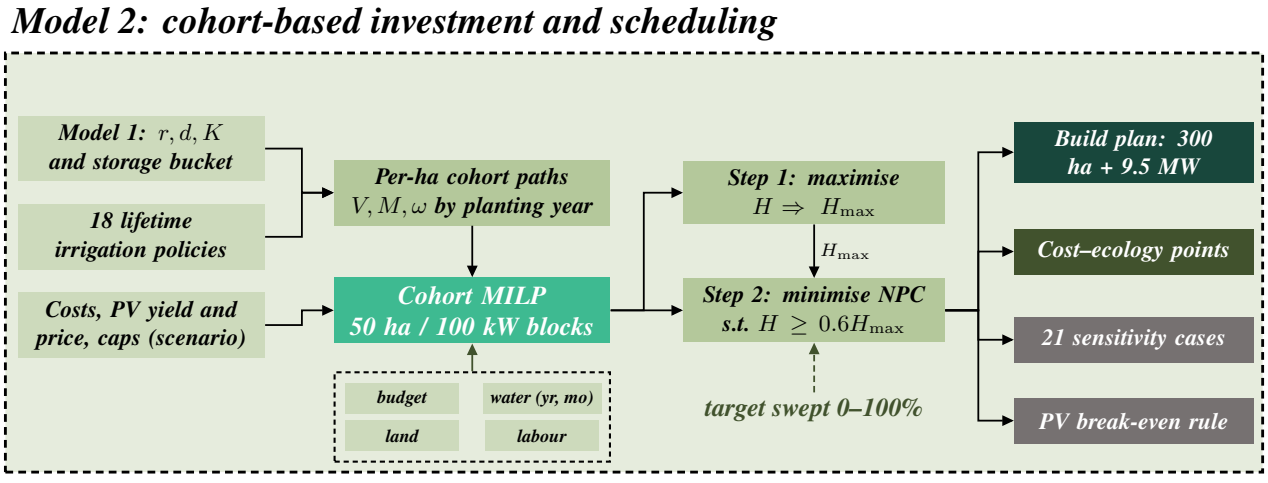


Figure 1: Structure of Model 2. Model 1 is run once per irrigation policy to get per-hectare paths; the optimiser then chooses how many 50 ha blocks of each (planting year, policy) cohort and how many 100 kW PV blocks to build.

1. **Cohorts.** A cohort $j = (s, p)$ is land planted in year s that follows lifetime irrigation policy p. All land of one cohort has the same state (V, M), so area only scales its water, cost, labour and ecology.
2. **Policy library (18 policies).** Constant 0, 100, 300, 600, 1,000 mm/yr; switch policies (100→300, 100→600, 300→600, 600→300, 1,000→600, 300→100) after age 3 or 5; and a ramp 100 (ages 1–3) → 300 (ages 4–7) → 600 mm/yr (age 8+). Rising and falling schedules are both allowed.
3. **Precompute.** Model 1 is run for each policy from $V_0 = 2\%$, $M_0 = 37.5$ mm, giving per-hectare V, M, ω (met-demand ratio) and water by age.
4. **MILP.** Because cohorts do not share soil water, every quantity is linear in the number of blocks: a mixed-integer linear programme over 15 years (SciPy `milp`/HiGHS, gap tolerance 0.2%, 30 s per solve).
5. **Two steps.** (i) maximise ecology H to get $H_{\max}$; (ii) minimise net present cost with $H \geq \varphi H_{\max}$, $\varphi = 0.6$. Sweeping $\varphi = 0, 0.2, \ldots, 1$ gives the cost–ecology points.

Timing within a year. Planting and PV construction are paid at the start of the year; upkeep, water and PV income at the end. New PV earns from the *next* year. Ecology in year t uses the cover at the start of that year. Ecology land and PV land do not overlap.

Writing note: Explain why cohorts are needed: a single "total area" state cannot tell a new stand from a mature one, and new stands need years to reach full cover and keep needing water.

3 Formulas and what they mean

3.1 Decisions and stocks

$$a_{sp} = \Delta_A n_{sp}, \quad y_s = \Delta_P z_s, \quad n_{sp}, z_s \in \mathbb{Z}_{\geq 0} \qquad \text{area and PV built, in blocks} \tag{1}$$

$$A_t = \textstyle\sum_{s \leq t, p} a_{sp}, \quad a_t = \textstyle\sum_p a_{tp} \qquad \text{area in service; new area} \tag{2}$$

$$P^{\text{inst}}_t = \textstyle\sum_{s \leq t} y_s, \quad P^{\text{op}}_t = \textstyle\sum_{s < t} y_s \qquad \text{PV installed; PV producing} \tag{3}$$

3.2 Link to Model 1 (each cohort)

$$(M_{sp,t+1},\ \omega_{sp,t}) = \mathcal{W}\big(V_{sp,t}, M_{sp,t}, q_{p,t-s}\big) \tag{4}$$

$$V_{sp,t+1} = V_{sp,t} + rV_{sp,t}(1 - V_{sp,t}/K)\,\omega_{sp,t} - dV_{sp,t} \tag{5}$$

$\mathcal{W}$ is the Model 1 storage bucket run over one year with quota $q_{p,h}$ (mm/yr) at stand age $h = t - s$; ω is the share of transpiration demand met. Parameters are those of Model 1 ($r = 0.547$, $d = 0.107$, $K = 38.08\%$). Model 1 has no wind-damage term (removed 2026-11-11; it was $k = 0$ in every Model 2 run, so no Model 2 number changes).

3.3 Ecology objective

$$H = \sum_t \sum_{s \leq t, p} a_{sp}\,\frac{V_{sp,t}}{K} \qquad \text{[effective ha} \cdot \text{yr]} \tag{6}$$

One hectare at full local cover ($V = K$) for one year counts as 1. It rewards both area and how green the area is, and it counts every year, so early planting is worth more.

3.4 Water, money, labour and PV

$$Q^e_t = 10 \sum_{s \le t,p} a_{sp}\, q_{p,t-s}, \qquad Q_t = Q^e_t + w_c P^{\mathrm{op}}_t \quad \text{gross water (m}^3\text{); 1 mm on 1 ha = 10 m}^3 \tag{7}$$

$$C^b_t = c_b a_t, \qquad C^p_t = c_p y_t \quad \text{planting and PV construction} \tag{8}$$

$$C^o_t = c_m A_t + c_{pm} P^{\mathrm{op}}_t + c_w Q_t \quad \text{upkeep of all stands, PV O\&M, water} \tag{9}$$

$$D_t = C^b_t + C^o_t + C^p_t, \qquad (f^b_t, f^o_t, f^p_t) = (C^b_t, C^o_t, C^p_t)/D_t \quad \text{spending and its three shares} \tag{10}$$

$$L_t = \ell_b a_t + \ell_m A_t + \ell_p y_t + \ell_{pm} P^{\mathrm{op}}_t \quad \text{labour (person-days)} \tag{11}$$

$$G_t = \sum_{s<t} y_s e_0 (1-d_p)^{t-s-1}, \qquad R_t = p_e G_t \quad \text{PV energy (kWh) and income} \tag{12}$$

Monthly water $Q_{t,m}$ splits each stand's quota by Model 1's monthly demand weights; PV cleaning water is spread evenly over 12 months.

3.5 Constraints

$$D_t \le B_t, \qquad \sum_t D_t \le B_{\mathrm{tot}} \quad \text{annual and total budget (income not added back)} \tag{13}$$

$$Q_t \le Q^{\max}, \qquad Q_{t,m} \le Q^{\max}_m \ (m = 1, \dots, 12) \quad \text{annual and monthly water} \tag{14}$$

$$A_t + \alpha_p P^{\mathrm{inst}}_t \le A^{\max}, \quad a_t \le a^{\max}, \quad P^{\mathrm{inst}}_t \le P^{\max} \quad \text{land, build rate, grid capacity} \tag{15}$$

$$L_t \le L^{\max} \quad \text{labour} \tag{16}$$

$$\underline{f}_p \sum_t D_t \le \sum_t C^p_t \le \overline{f}_p \sum_t D_t \quad \text{optional PV share of total spending} \tag{17}$$

3.6 Cost objective and trade-off

$$N = \sum_{t=0}^{T-1} \Big[\delta^t \big(C^b_t + C^p_t\big) + \delta^{t+1} \big(C^o_t - R_t\big) \Big], \qquad \delta = \frac{1}{1+0.05} \tag{18}$$

$$\text{Step 1: } H_{\max} = \max H \quad \text{s.t. constraints;} \qquad \text{Step 2: } \min N \quad \text{s.t. } H \ge \varphi H_{\max}, \text{ constraints.} \tag{19}$$

The target share φ is the explicit weight between ecology and cost (0.6 in the base case). Ecology is not converted into money. All sensitivity runs keep the same absolute target (1,258.22 effective ha·yr), so no case looks cheaper by doing less.

3.7 PV break-even rule

$$p_e \;\ge\; p^{\mathrm{BE}}_e(s) = \frac{c_p + \sum_{t=s+1}^{T-1} \delta^{t+1-s} \left(c_{pm} + c_w w_c\right)}{\sum_{t=s+1}^{T-1} \delta^{t+1-s}\, e_0 (1-d_p)^{t-s-1}} \tag{20}$$

Price per kWh at which 1 kW built in year s pays back its own cost within the horizon. Necessary for a standalone PV block, not sufficient for the whole plan (it ignores competition for budget, land and water).

4 Algorithms

Model 2 is solved in three stages: build the problem from Model 1 (Alg. 1), solve it with the two-step ε-constraint method (Alg. 2), and check every answer by replaying it through Model 1 (Alg. 3). Formulas refer to Section 3 and to Section 4 of `Reports/model_v7/model_v7.tex`.

Algorithm 1 Build the cohort MILP

Input: scenario parameters (Table 2); Model 1 r, d, K, bucket parameters, monthly climatology; policy library $\mathcal{P}$ (18 policies)

Output: matrix $\boldsymbol{A}$, limits $\boldsymbol{b}$, ecology vector $\boldsymbol{h}$, cost vector $\boldsymbol{n}$, integer upper bounds

for each policy $p \in \mathcal{P}$ **do**
  Run Model 1 (Alg. 3 of the M1 brief, year after year) for T years from $V_0 = 2\%$, $M_0 = 37.5$ mm
  Store per hectare and by stand age: V_{start}, irrigation (mm), monthly gross irrigation
for each start year $s = 1, \ldots, T$ **do**
  for each policy p **do** ▷ drop policies identical over the remaining $T - s + 1$ years
    Column "ecology (s, p)" = one 50 ha block; fill year by year: new area, area in service, land, planting cost, upkeep + water cost, water (yearly, monthly), labour, effective ha = $50\, V_{\text{start}}/K$
  if $s < T$ **then** ▷ PV built in the last year earns nothing
    Column "PV s" = one 100 kW block: capex in year s; from $s + 1$: energy $e_0(1 - d_p)^{\text{age}}$, income, O&M, cleaning water, labour; land from s
$\boldsymbol{n} \leftarrow$ discounted cost per column (capex at start of year, O&M and income at end), Eq. (18)
$\boldsymbol{h} \leftarrow$ sum over years of effective ha per column, Eq. (6)
Rows of $\boldsymbol{A}\boldsymbol{z} \le \boldsymbol{b}$: total budget; PV share bounds; for each year: budget, water, land, new area, labour, PV capacity; for each year and month: water
Integer bounds: ecology $\le \lfloor \min(200, 1200)/50 \rfloor$ blocks, PV $\le 10{,}000/100$ blocks

Code: `Code/v7_m2_core.py` (`policy_library`, `simulate_policy`, `build_problem`).

Algorithm 2 Two-step ε-constraint solution, trade-off points and sensitivity

Input: problem from Alg. 1; target share $\varphi = 0.6$; gap 0.2%; time limit 30 s per solve

Output: base plan, $H_{\max}$, six trade-off points, 21 sensitivity plans

Step 1: $z^{\max} \leftarrow \arg\max\ \boldsymbol{h}^\top z$ s.t. $\boldsymbol{A}z \le \boldsymbol{b}$, $z \in \mathbb{Z}_{\ge 0}$; $\quad H_{\max} \leftarrow \boldsymbol{h}^\top z^{\max}$
Step 2: $z^* \leftarrow \arg\min\ \boldsymbol{n}^\top z$ s.t. $\boldsymbol{A}z \le \boldsymbol{b}$, $\boldsymbol{h}^\top z \ge \varphi H_{\max}$ ▷ base plan
for each solve **do**
  Call the MILP solver (HiGHS through SciPy `milp`); record status, objective, bound and gap
  If no feasible plan: report "infeasible" and stop for this case (no made-up answer)
  Round z and check integrality (10^{-4}) and every constraint (10^{-7}) independently
for $\varphi \in \{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ **do**
  Solve Step 2 with target $\varphi H_{\max}$ ▷ Table 8
for each sensitivity case (one parameter changed) **do**
  Rebuild the problem (Alg. 1); solve Step 2 with the *same* target $0.6 H_{\max} = 1{,}258.22$ ▷ Table 9
For each build year s: compute the PV break-even price, Eq. (20), and standalone NPV ▷ Table 10

Code: `Code/v7_m2_core.py` (`solve`), `Code/v7_model2.py` (`run`).

Algorithm 3 Replay check of a solved plan

Input: chosen block counts z^*

Output: yearly and monthly ledgers; largest difference from the MILP rows

for each chosen ecology cohort (s, p) with area a **do**
  Re-run Model 1 for policy p from year s to T (not the cached columns)
  Add its area, water (yearly, monthly), upkeep, labour, effective ha to the yearly ledger
for each chosen PV block **do**
  Add capex, energy, income, O&M, cleaning water, land, labour by year
Compute fund shares (f_t^b, f_t^o, f_t^p), budget left, discounted net cost
Compare every ledger column with $\boldsymbol{A}z^*$, $\boldsymbol{h}^\top z^*$, $\boldsymbol{n}^\top z^*$; base case largest difference 1.9×10^{-9}

Code: `Code/v7_m2_core.py` (`replay`). Run: `python Code/v7_model2.py`; tests: `python -m unittest`

`discover -s Code -p test_v7_m2.py` (7 tests, incl. brute-force check on a small case).

Writing note: Explain in the paper why the problem is linear although Model 1 is not: Model 1 is run once per policy before optimising, and cohorts do not share soil water, so every cost, water and ecology term is the per-block value times the number of blocks.

5 Parameters

Parameter	Base value	Source / status
Horizon T	15 years	within the recommended 10–20 years
Blocks Δ_A, Δ_P	50 ha; 100 kW	numerical grid (25 ha also tested)
New stand V_0, M_0	2%; 37.5 mm	assumption (1% and 4% tested)
Total / annual budget B_{tot}, B_t	80 M; 12 M/yr	planning scenario
Land $A^{\max}$; build rate $a^{\max}$	1,200 ha; 200 ha/yr	planning scenario
Water $Q^{\max}$	1.5 M m^3/yr	scenario, not a water permit
Monthly water $Q_m^{\max}$ (Jan–Dec)	50, 60, 90, 120, 180, 220, 240, 220, 150, 90, 50, 30 thousand m^3	scenario delivery capacity
Labour $L^{\max}$	12,000 person-days/yr	scenario
Planting cost c_b	52,500 CNY/ha	Yutian 2024 desert-planting trial (3.5 M for 1,000 mu), transferred; ±25% tested
Upkeep c_m	1,500 CNY/(ha·yr)	assumption; ±50% tested
Water price c_w	1 CNY/m^3	assumption
Labour rates ℓ_b, ℓ_m	30 person-days/ha; 3 per ha·yr	assumption
PV capex c_p; O&M c_{pm}	3,000 CNY/kW; 50 CNY/(kW·yr)	assumption; 2,250 / 3,750 tested
PV net yield e_0	1,500 kWh/(kW·yr)	assumption, net of losses
PV price p_e	0.30 CNY/kWh	scenario; 0.15 / 0.40 tested
PV degradation d_p	0.5%/yr	Jordan & Kurtz (2013) median
PV land α_p; labour ℓ_p, ℓ_{pm}	0.002 ha/kW; 0.10 and 0.01 person-days/kW	assumption
PV cleaning water w_c	0.02 m^3/(kW·yr)	assumption
PV capacity $P^{\max}$	10,000 kW	scenario grid limit
Discount rate	5%/yr (real)	scenario
Ecological target φ	0.60 of $H_{\max}$	policy choice for the demonstration
Model 1 inputs	r = 0.547, d = 0.107, K = 38.08%, C = 75 mm	from Model 1

Table 2: Model 2 parameters. The problem gives no numbers for budget, prices or caps, so all of these are stated assumptions.

6 Results

6.1 Maximum ecology reference

With every resource devoted to ecology: $H_{\max}$ = 2,097.03 effective ha·yr (upper bound 2,098.93, gap 0.09%), 900 ha planted, no PV, all 80 M spent, net present cost 64.42 M. The base target is $0.6H_{\max}$ = 1,258.22.

6.2 Optimal base plan

Cohort / asset	Policy (mm/yr by stand age)	Year built	Size
Ecology A	100 (ages 1–3) → 300	1	100 ha
Ecology B	ramp 100 (1–3) → 300 (4–7) → 600 (8+)	1	50 ha
Ecology C	100 (ages 1–3) → 300	2	150 ha
PV	–	1 / 2 / 3 / 4	1,200 / 1,100 / 3,700 / 3,500 kW

Table 3: What to build. Total 300 ha and 9,500 kW.

Indicator	Value	Note
Ecology achieved	1,264.84 eff. ha·yr	target 1,258.22
Total spending (undiscounted)	68.93 M	86% of 80 M
planting / upkeep + water / PV construction	15.75 / 24.68 / 28.50 M	
PV income (undiscounted, 15 yr)	49.90 M	
Net present cost N	25.35 M	lower bound 25.30 M, gap 0.20%
Without PV (best plan)	27.84 M	PV saves 2.49 M (9%)
Peak annual water	1.050 M m^3	70% of annual cap
Peak monthly water use	94.9% of May cap	years 8–15
Peak labour	5,522 person-days	46% of cap
Land used	319 ha	of 1,200 ha

Table 4: Base plan summary.

Year	New ha	Stock ha	New PV kW	PV producing kW	Planting M	Upkeep+water M	PV build M	Spending M	PV income M
1	150	150	1,200	0	7.88	0.38	3.60	11.85	0.00
2	150	300	1,100	1,200	7.88	0.81	3.30	11.99	0.54
3	0	300	3,700	2,300	0.00	0.87	11.10	11.97	1.03
4	0	300	3,500	6,000	0.00	1.35	10.50	11.85	2.69
5	0	300	0	9,500	0.00	1.83	0.00	1.83	4.25
6	0	300	0	9,500	0.00	1.83	0.00	1.83	4.23
7	0	300	0	9,500	0.00	1.83	0.00	1.83	4.21
8	0	300	0	9,500	0.00	1.98	0.00	1.98	4.19
9	0	300	0	9,500	0.00	1.98	0.00	1.98	4.17
10	0	300	0	9,500	0.00	1.98	0.00	1.98	4.15
11	0	300	0	9,500	0.00	1.98	0.00	1.98	4.13
12	0	300	0	9,500	0.00	1.98	0.00	1.98	4.11
13	0	300	0	9,500	0.00	1.98	0.00	1.98	4.09
14	0	300	0	9,500	0.00	1.98	0.00	1.98	4.07
15	0	300	0	9,500	0.00	1.98	0.00	1.98	4.05

Table 5: Base plan, year by year: building and money.

Year	Water (k m^3)	Labour (p-d)	Effective ha	Shares build / O&M / PV	Budget left M	Discounted net cost M
1	150.0	5,070	7.9	0.66 / 0.03 / 0.30	68.15	11.83
2	300.0	5,522	19.0	0.66 / 0.07 / 0.28	56.16	10.89
3	300.0	1,293	26.7	0.00 / 0.07 / 0.93	44.20	9.92
4	600.1	1,310	35.2	0.00 / 0.11 / 0.89	32.35	7.97
5	900.2	995	46.2	0.00 / 1.00 / 0.00	30.52	−1.90
6	900.2	995	62.3	0.00 / 1.00 / 0.00	28.70	−1.80
7	900.2	995	80.5	0.00 / 1.00 / 0.00	26.87	−1.70
8	1,050.2	995	95.1	0.00 / 1.00 / 0.00	24.90	−1.50
9	1,050.2	995	107.5	0.00 / 1.00 / 0.00	22.92	−1.41
10	1,050.2	995	117.2	0.00 / 1.00 / 0.00	20.95	−1.33
11	1,050.2	995	124.6	0.00 / 1.00 / 0.00	18.97	−1.26
12	1,050.2	995	130.2	0.00 / 1.00 / 0.00	17.00	−1.19
13	1,050.2	995	134.5	0.00 / 1.00 / 0.00	15.02	−1.12
14	1,050.2	995	137.8	0.00 / 1.00 / 0.00	13.05	−1.06
15	1,050.2	995	140.3	0.00 / 1.00 / 0.00	11.07	−1.00

Table 6: Base plan, year by year: resources, fund shares, ecology. Effective ha = $\sum$ area $\times V/K$; their sum over 15 years is H.

- **Front-loaded.** Planting and PV fill the 12 M annual budget in years 1–4 (slack 0.15, 0.015, 0.035, 0.15 M); from year 5 only upkeep is paid.
- **PV pays for upkeep.** From year 5, PV income (4.05–4.25 M/yr) exceeds all upkeep and water (1.83–1.98 M/yr), so the yearly net cost is negative.
- **Water steps up with stand age**: 150 k m^3 (year 1) → 300 k → 600 k (year 4) → 900 k (year 5) → 1,050 k m^3 (year 8 on).
- **Monthly bottleneck.** Highest monthly use relative to cap: May 94.9%, April 88.7%, June 87.3%, July 86.1%; the annual total uses only 70%.

Cohort	Cover V at start of project year (%) [quota mm/yr]					
	1	3	5	8	10	15
A (100 ha, yr 1)	2.00 [100]	3.95 [100]	6.75 [300]	12.76 [300]	14.68 [300]	16.90 [300]
B (50 ha, yr 1, ramp)	2.00 [100]	3.95 [100]	6.75 [300]	12.76 [600]	18.41 [600]	23.15 [600]
C (150 ha, yr 2)	–	2.82 [100]	4.98 [300]	11.37 [300]	13.83 [300]	16.64 [300]

Table 7: Cohort trajectories inherited from Model 1 (K = 38.08%).

6.3 Ecology–cost trade-off

Target φ	H achieved	Net present cost M	Spending M	Area ha	PV kW	Marginal cost (k CNY per eff. ha·yr)
0%	0	−4.91	36.60	0	10,000	–
20%	428.05	4.78	49.22	100	10,000	22.6
40%	839.97	14.17	58.72	150	10,000	22.8
60%	1,264.84	25.35	68.93	300	9,500	26.3
80%	1,678.79	40.52	70.59	500	5,300	36.7
100%	2,097.03	64.42	80.00	900	0	57.1

Table 8: Six solved points (not a fitted curve). Marginal cost = change in N / change in H from the previous row.

- Cost rises faster than ecology: the marginal cost per effective ha·yr is flat to 40%, then rises 2.5-fold by 100%.
- As the target rises, PV is pushed out (10 MW → 0) because budget is needed for planting and upkeep.
- With no ecological requirement, PV alone earns 4.91 M in present value; this is not a reason to skip treatment.

6.4 Sensitivity (same absolute target 1,258.22)

Case	N (M)	Change vs base	Area ha	PV kW	MIP gap %
Base	25.35	–	300	9,500	0.20
No PV allowed	27.84	+9.8%	300	0	0.04
PV price 0.15 CNY/kWh	27.84	+9.8%	300	0	0.04
PV price 0.40 CNY/kWh	14.28	−43.7%	300	10,000	0.10
PV capex 2,250 CNY/kW	17.76	−30.0%	300	10,000	0.00
PV capex 3,750 CNY/kW	27.84	+9.8%	300	0	0.04
PV share of spending ≥ 10%	25.35	0.0%	300	9,500	0.20
PV share of spending ≥ 30%	25.35	0.0%	300	9,500	0.20
O&M −50% (750 /ha, 25 /kW)	21.21	−16.3%	300	10,000	0.17
O&M +50% (2,250 /ha, 75 /kW)	29.09	+14.8%	300	5,600	0.13
Planting cost 39,375 CNY/ha	20.86	−17.7%	300	10,000	0.08
Planting cost 65,625 CNY/ha	29.93	+18.1%	300	8,000	0.18
Water caps −20%	27.82	+9.7%	350	8,500	0.20
Water caps +10%	24.99	−1.4%	300	9,500	0.20
Total budget 64 M	25.45	+0.4%	300	8,100	0.14
Labour cap 6,000	25.35	0.0%	300	9,500	0.13
Initial cover 1%	32.74	+29.1%	400	7,500	0.19
Initial cover 4%	20.42	−19.5%	200	10,000	0.19
Area block 25 ha	25.29	−0.2%	300	9,400	0.20
Storage capacity 37.5 mm	25.79	+1.7%	300	9,300	0.20
Storage capacity 150 mm	25.14	−0.8%	300	9,400	0.20
Constant-quota policies only	27.04	+6.6%	300	8,800	0.00

Table 9: One-at-a-time sensitivity. Differences below about 0.4% are within solver tolerance.

- **PV is all-or-nothing on price.** At 0.15 CNY/kWh or 3,750 CNY/kW no PV is built; at 0.40 CNY/kWh or 2,250 CNY/kW the full 10 MW is built and cost falls 30–44%.
- **Establishment success matters most among ecology inputs**: initial cover 1% vs 4% moves cost by +29% / −20% and needs 400 vs 200 ha.
- Upkeep and planting prices move cost by 15–18% for ±25–50%.
- Labour and the budget are not binding at this target; water binds when caps are cut by 20% (the plan changes to 350 ha and 8.5 MW, +9.7%).
- Allowing rising/falling schedules saves 6.6% versus constant quotas only.

6.5 PV screening by construction year

Build year	1	2	3	4	5	6	8	10
Break-even price (CNY/kWh)	0.253	0.264	0.277	0.293	0.312	0.335	0.402	0.523
Standalone NPV (CNY/kW)	652	473	284	85	−126	−348	−830	−1,368

Table 10: At the base price 0.30 CNY/kWh, PV pays for itself only if built in years 1–4, which are exactly the years the optimiser chose.

6.6 Verification

29 optimisation runs, all feasible, all gaps ≤ 0.20%. Every plan is replayed through Model 1 and every ledger (money, water, labour, land, PV) is recomputed: largest difference 1.9×10^{-9}. 7 unit tests, including a brute-force check of all integer plans on a small instance against the MILP answer.

7 Actionable process criteria

1. **PV entry rule.** Build a PV block in year s only if the expected price satisfies Eq. (20), $p_e \geq p_e^{\mathrm{BE}}(s)$ (base: 0.253 in year 1 rising to 0.312 in year 5), *and* the plan still meets the budget, monthly water and land limits. Under base prices this means PV only in years 1–4.
2. **Water follows stand age.** Give new stands about 100 mm/yr (1,000 m^3/ha) for the first three years and raise to 300 mm/yr from year 4; a share of the area can go to 600 mm/yr from year 8 to lift cover further. Young stands use little water: in Model 1, quotas above about 300 mm/yr give no extra growth in the first five years (the surplus drains), and the optimiser found 100 mm/yr cheapest for the first three.
3. **Plant early.** Put all planting in the first two years (as fast as the annual budget allows), since every year of growing cover adds to the ecological total and stands need 10–15 years to approach their plateau.
4. **Check months, not only years.** Schedule water against the May, April and June caps; they bind long before the annual total.
5. **Choose the target where the cost curve bends.** Up to 40–60% of the maximum, each effective ha·yr costs 23–26 k CNY; above 80% it costs 57 k CNY. Raise the target only if that price is acceptable.
6. **Measure establishment first.** Initial cover is the input that changes cost most; a pilot survival survey is the most valuable data to collect.

8 What we can claim and what we cannot

Can claim	Must not claim
A dynamic plan that tracks every cohort's vegetation and soil water and respects budget, water (yearly and monthly), land and labour.	That 300 ha and 9.5 MW is a recommended project size; budget, caps and prices are assumptions.
Optimality within the 18-policy library and 50 ha grid, with gap ≤ 0.2%.	Global optimum over all possible watering schedules or continuous areas.
PV is worth building only early and only above the break-even price; it can finance upkeep.	That more PV is always better, or a guaranteed PV income.
Rising marginal cost of ecology; the monthly water cap is the bottleneck.	That H is a measured ecological benefit, erosion reduction or money value.

9 Figures ready for the paper

File (`Figures/M2/`)	What it shows / caption point
`m2_f01_build_and_spending`	New and stock area; PV built and producing; three spending streams vs annual budget; spending shares
`m2_f02_cohorts`	Cover, quota and storage of the three cohorts; their contribution to effective area
`m2_f03_water_pressure`	Annual and monthly water vs caps; month-by-year use of the cap
`m2_f04_cost_ecology_points`	Six solved trade-off points; area, PV and MIP gap at each
`m2_f05_investment_sensitivity`	Cost, PV and spending mix under price, cost and PV-share cases
`m2_f06_assumption_checks`	Initial cover, storage, caps, grid, constant-only policies
`m2_f07_pv_screening`	Break-even price and standalone NPV by build year

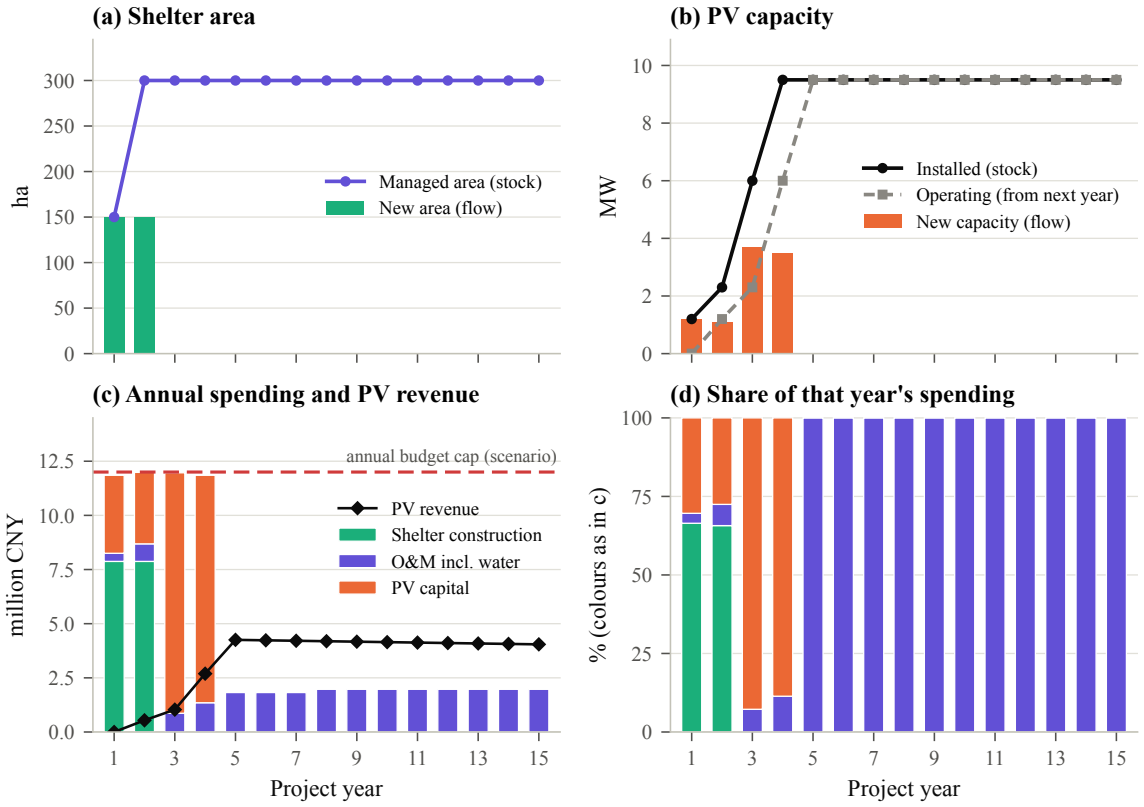


Figure 2: (From `m2_f01`) Base plan: building, PV and spending by year.

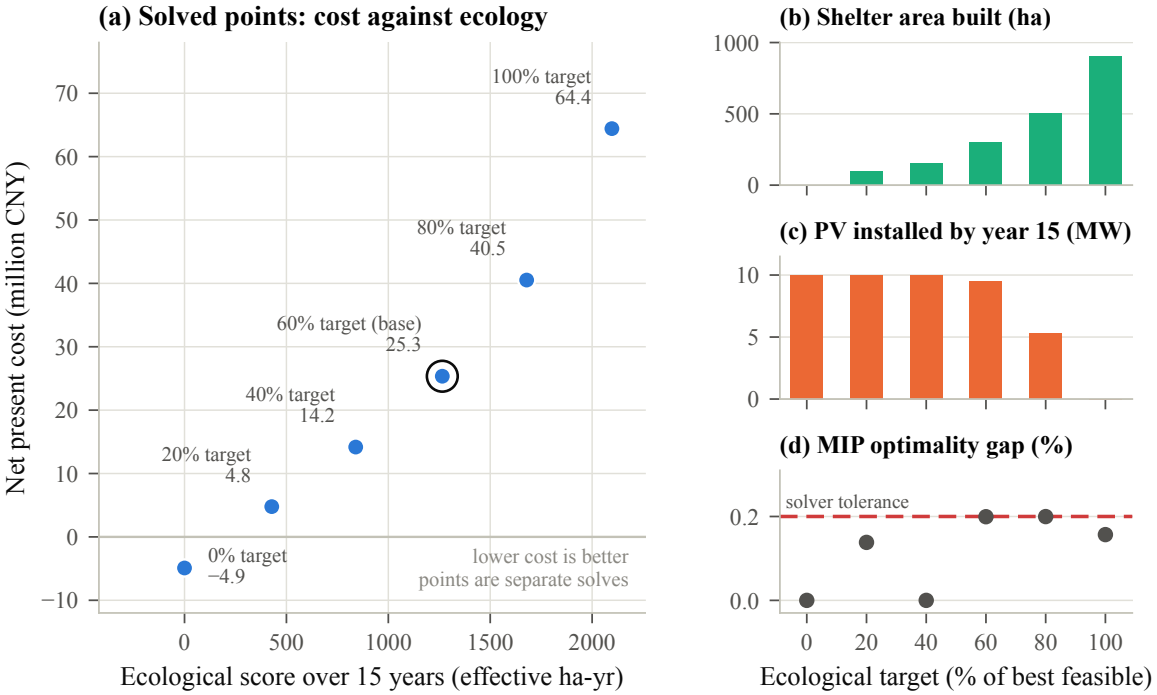


Figure 3: (From `m2_f04`) Ecology–cost trade-off at six solved targets.

10 Suggested outline for the Model 2 section

1. Why cohorts: link to Model 1 states; Fig. 1.
2. Decision variables (area, water, three fund shares, labour) and Eqs. (1)–(12).
3. Constraints (13)–(16) and the ε-constraint objective (18)–(19).
4. Algorithms: build (Alg. 1), two-step solve (Alg. 2), replay check (Alg. 3).
5. Parameters (Table 2) with the note that they are scenario values.
6. Base plan (Tables 3–6) and the trade-off (Table 8).
7. Sensitivity of the investment structure (Table 9) and PV screening (Table 10).
8. Actionable criteria (Section 7) and limits.